

Installing Arch Linux with BTRFS and Disk Encryption

 itsfoss.com/arch-linux-install-encrypted-btrfs

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On our [Arch installation video](#), a viewer requested a tutorial on installing Arch but with BTRFS and with encryption enabled.

And hence this tutorial came into existence.

I am using the official `archinstall` script. Though a command line tool, this guided installer allows even a moderate system user to enjoy the "greatness" of Arch Linux.

The method discussed here **wipes out the existing operating system(s)** from your computer and installs Arch Linux on it. So if you are going to follow this tutorial, make sure that you have backed up your files externally, or else you'll lose all of them. You have been warned!

Requirements

Here's what I recommend for this tutorial:

- An x86_64 (i.e. 64 bit) compatible machine
- Minimum 2 GB of RAM (recommended 4-8 GB, depending upon the desktop environment or window manager you choose)
- At least 10 GB of free disk space (recommended 20 GB for basic usage with a desktop environment)

- An active internet connection
- A USB drive with a minimum 4 GB of storage capacity
- Familiarity with the Linux command line

Once you have made sure you have all the requirements, let's install Arch Linux.

Step 1: Download the Arch Linux ISO

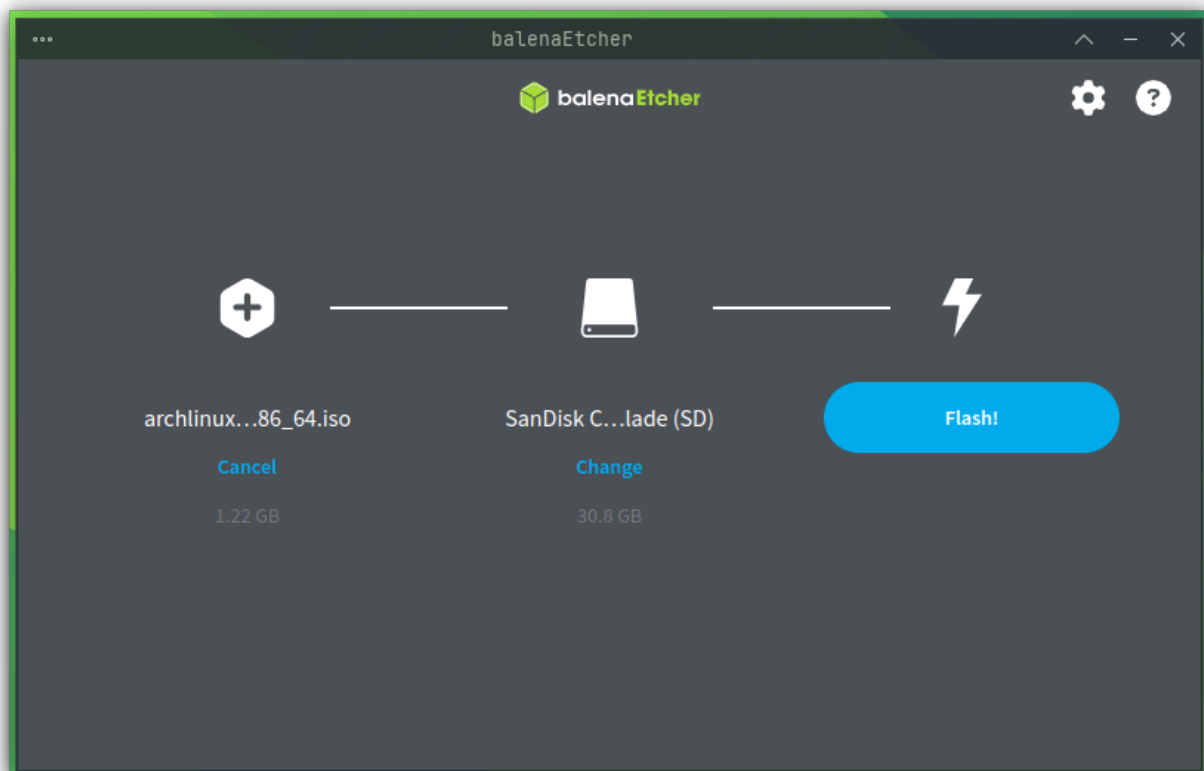
Download the ISO from the official website. Both direct download and torrent links are available.

[Download Arch Linux](#)

Step 2: Create a live USB of Arch Linux

You will have to create a live USB of Arch Linux from the ISO you just downloaded.

You may use the [Etcher GUI tool](#) to create the live USB. It is available for both Windows and Linux.



Etcher Live USB creation

Alternatively, if you are on Linux, you can [use the dd command to create a live USB](#). Replace `/path/to/archlinux.iso` with the path where you have downloaded the ISO file, and `/dev/sdx` with your USB drive in the example below. You can get your drive information using [lsblk](#) command.

```
dd bs=4M if=/path/to/archlinux.iso of=/dev/sdx status=progress && sync
```

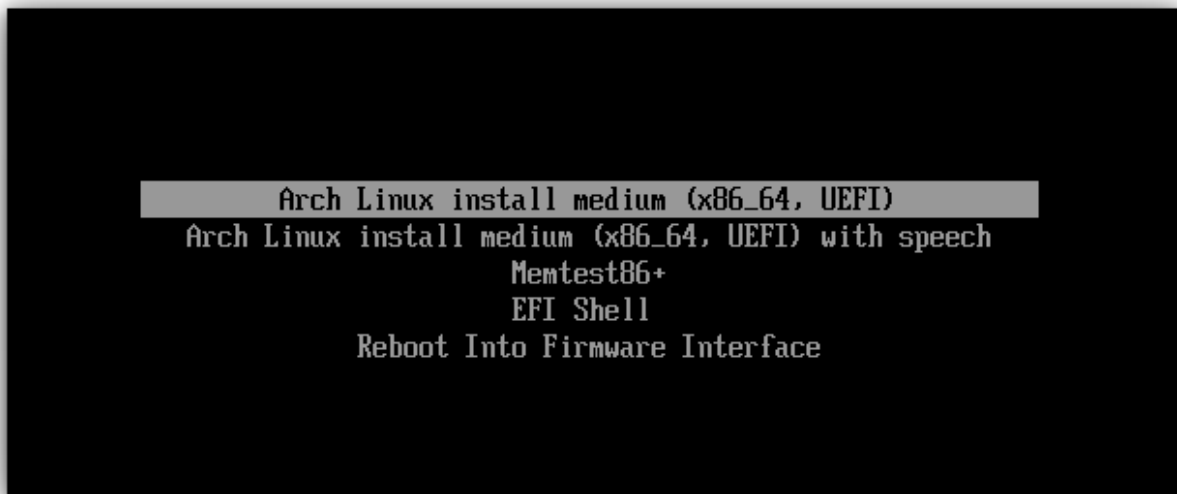
Basically, choose any live USB creation tool you like.

Step 3: Boot from the live USB

Do note that in some cases, you may not be able to boot from live USB with secure boot enabled. If that's the case with you, [disable secure boot first](#).

Once you have created a live USB for Arch Linux, shut down your PC. Plug in your USB and boot your system. While booting, keep pressing F2, F10 or F12 key (depending upon your system) to [access UEFI boot settings](#).

Here, select to boot from USB or removable disk. Once you do that and the system boots, you should see an option like this:



UEFI Boot Screen

Select Arch Linux UEFI (x86_64) option to start the live medium.

Legacy BIOS users should select the x86_64 BIOS option.

Step 4: Connect to Wi-Fi

You need an active internet connection for [installing Arch Linux](#).

If you have wired connection, good. Else, you need to make some effort to connect to your Wi-Fi before starting the archinstall script.

First, in the Arch Linux live prompt, enter the command:

```
iwctl
```

This Internet Wireless daemon control is used to enrol Wi-Fi connection to your system. As soon as you enter the command, you can see that the prompt has changed to `iwd`.

Here, you need to list devices to get the name of your wireless hardware device.

```
root@archiso ~ # iwctl
NetworkConfigurationEnabled: disabled
StateDirectory: /var/lib/iwd
Version: 3.2
[iwd]# device list
```

Devices				
Name	Address	Powered	Adapter	Mode
wlan0		on	phy0	station

List network devices

In the above screenshot, you can see the name of my Wi-Fi device is `wlan0`.

Now, use this device to scan available Wi-Fi connections in the vicinity.

```
station wlan0 scan
station wlan0 get-networks
```

```
[iwd]# station wlan0 scan
[iwd]# station wlan0 get-networks
```

Available networks		
Network name	Security	Signal
> FTTH-91BB	psk	*****
	psk	*****

```
[iwd]# station wlan0 connect FTTH-91BB
Type the network passphrase for FTTH-91BB psk.
Passphrase: *****
[iwd]#
```

Connect to a Wi-Fi

This will print the name of the Wi-Fi services available. Note the “Network Name”.

To connect to the network, use the command:

```
station wlan0 connect "Network Name"
```

This will ask you to enter the Wi-Fi password. Enter it and you should be now connected to internet.

Exit the `iwd` prompt using CTRL+D.

You can check if the network is functioning [using the ping command](#):

```
ping google.com
```

```
root@archiso ~ # ping google.com
PING google.com (142.250.195.238) 56(84) bytes of data:
64 bytes from maa03s43-in-f14.1e100.net (142.250.195.238): icmp_seq=1 ttl=255 time=16.5 ms
64 bytes from maa03s43-in-f14.1e100.net (142.250.195.238): icmp_seq=2 ttl=255 time=16.9 ms
64 bytes from maa03s43-in-f14.1e100.net (142.250.195.238): icmp_seq=3 ttl=255 time=17.1 ms
64 bytes from maa03s43-in-f14.1e100.net (142.250.195.238): icmp_seq=4 ttl=255 time=16.9 ms
64 bytes from maa03s43-in-f14.1e100.net (142.250.195.238): icmp_seq=5 ttl=255 time=17.2 ms
^C
--- google.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4315ms
rtt min/avg/max/mdev = 16.512/16.927/17.224/0.241 ms
root@archiso ~ #
```

Ping Google

Step 5: Pacman download settings

Before starting the archinstall script, let's change the download limit of pacman.

Edit the pacman configuration using:

```
nano /etc/pacman.conf
```

Here, uncomment the `parallelDownload` option and set a value according to your internet speed.

If you have a decent internet speed, set the parallel download count to 10.

On my test system, I needed to run `pacman -Sy` and then `pacman -S archlinux-keyring` (install Arch Linux keyrings) before starting the installer. Otherwise, the installer crashed with some errors. You may also need to read carefully what the prompt error says.

Step 6: Start Archinstall script

With the network connection ready, let's start the archinstall script with the command below:

```
archinstall
```

This will start the text-based arch installation script.

```
Press ? for help
> Archinstall language      English (100%)
  Locales                  +
  Mirrors
  Disk configuration
  Swap                    +
  Bootloader              +
  Unified kernel images
  Hostname                +
  Root password
  User account
  Profile
  Audio
  Kernels                  +
  Network configuration
  Additional packages
  Optional repositories
  Timezone                +
  Automatic time sync (NTP) +

  Save configuration
  Install
  Abort
```

Archinstall script

Set the installation language

The first setting in the installer is the installation language. This option sets what language is used in the Terminal User Interface.

The latest archinstall provides a percentage value corresponding to each language, that describes how much translation has been completed.

NOTE: If a language can not displayed properly, a proper font must be set manually in the console.
All available fonts can be found in "/usr/share/kbd/consolefonts"
e.g. setfont LatGrkCyr-8x16 (to display latin/greek/cyrillic characters)

```
Select language
Arabic (100%)
Brazilian Portuguese (88%)
Catalan (100%)
Czech (100%)
Dutch (35%)
> English (100%)
Estonian (74%)
Finnish (83%)
French (100%)
Georgian (88%)
German (81%)
Hebrew (81%)
Hindi (3%)
Hungarian (100%)
Indonesian (54%)
Irish (88%)
Italian (83%)
Japanese (100%)
Korean (54%)
Lithuanian (19%)
Modern Greek (1453-) (61%)
Polish (67%)
Portuguese (100%)
Romanian (79%)
Russian (88%)
Simplified Chinese (80%)
Spanish (99%)
Swedish (100%)
Tamil (88%)
Traditional Chinese (71%)
Turkish (84%)
Ukrainian (56%)
Urdu (31%)
```

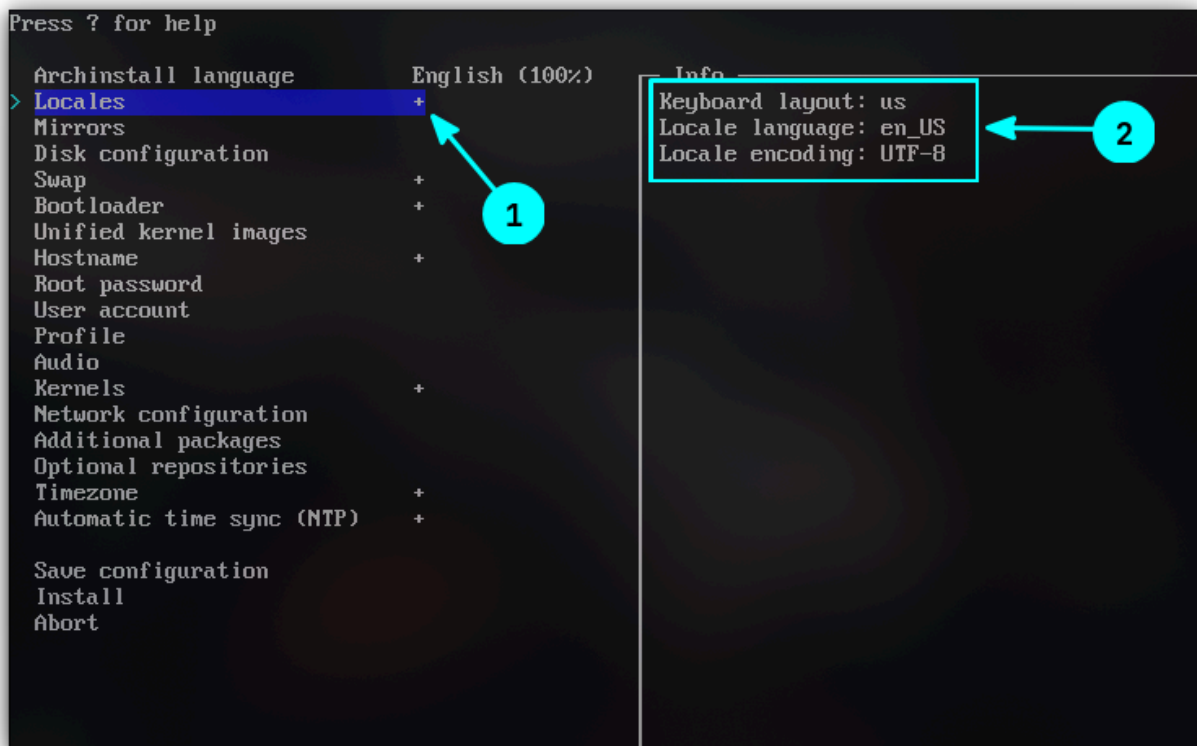
Installation language

I will be going with the default English.

Locale Settings

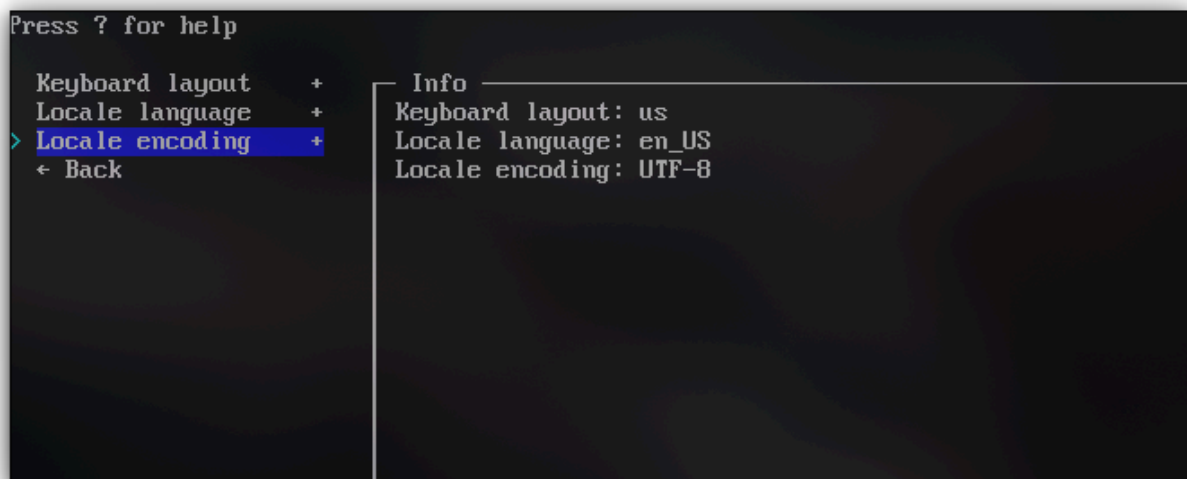
You should set your locale and keyboard settings. Here, if you are OK with the defaults, you can skip to the next setting.

Some programs like Rofi launcher may not launch if your locale is different from en_US. So, adding en_US as a locale is a good thing to avoid future headaches.



Set keyboard and locale settings

To change a setting, press the enter key to go inside and select individual items.



Inside locale settings

Mirror settings

Press the enter key on the **Mirrors** in the main menu of archinstall script. This will bring you to the mirror selection section.

Enter the Mirror Region.

```
Press ? for help
> Mirror region ←
  Custom mirrors
  ← Back
```

Select the Mirror Region option.

This will provide a list of countries. You can select a country near your location for a faster network.

```
Press ? for help

Mirror regions
> [x] India
  [ ] Indonesia
  [ ] Iran
  [ ] Israel
  [ ] Italy
[x] Japan
  [ ] Kazakhstan
  [ ] Kenya
  [ ] Latvia
  [ ] Lithuania
  [ ] Mexico
  [ ] Moldova
  [ ] Nepal
  [ ] Netherlands
  [ ] New Caledonia
  [ ] New Zealand
  [ ] North Macedonia
  [ ] Norway
  [ ] Poland
  [ ] Portugal
  [ ] Romania
  [ ] Russia
  [ ] Réunion
  [ ] Saudi Arabia
  [ ] Serbia
[x] Singapore
  [ ] Slovakia
  [ ] Slovenia
  [ ] South Africa
  [ ] South Korea
  [ ] Spain
  [ ] Sweden
  [ ] Switzerland
  [ ] Taiwan
  [ ] Thailand
  [ ] Türkiye
  [ ] Ukraine
  [ ] United Arab Emirates
  [ ] United Kingdom
  [ ] United States
  [ ] Uzbekistan
  [ ] Vietnam
  [ ] Worldwide
```

Mirror Countries (Click to expand the image)

Use the "/" key to start a search. TAB key to select/mark an entry. Once multiple entries are marked, use the ENTER key to set those countries as mirrors.

The mirrors from selected countries will be listed. Move to "Back" and click enter.

```
Press ? for help
> Mirror region +
  Custom mirrors
  ← Back

Info
Australia
-----
https://mirror.aarnet.edu.au/pub/archlinux/
http://ftp.iinet.net.au/pub/archlinux/
http://mirror.internode.on.net/pub/archlinux/
http://ftp.swin.edu.au/archlinux/
http://syd.mirror.rackspace.com/archlinux/
https://syd.mirror.rackspace.com/archlinux/
http://archlinux.mirror.digitalpacific.com.au/
https://archlinux.mirror.digitalpacific.com.au/
http://au.mirrors.cicku.me/archlinux/
https://au.mirrors.cicku.me/archlinux/
http://gsl-syd.mm.fcix.net/archlinux/
https://gsl-syd.mm.fcix.net/archlinux/
https://sydney.mirror.pkgbuild.com/
https://au.arch.niranjana.co/

India
-----
http://in.mirrors.cicku.me/archlinux/
https://in.mirrors.cicku.me/archlinux/
https://mirror.albony.in/archlinux/
https://mirror.maa.albony.in/archlinux/
https://mirror.nag.albony.in/archlinux/
http://in-mirror.garudalinux.org/archlinux/
https://in-mirror.garudalinux.org/archlinux/
http://mirror.sahil.world/archlinux/
https://mirror.sahil.world/archlinux/
https://mirrors.abhy.me/archlinux/
http://in.arch.niranjana.co/
https://in.arch.niranjana.co/
http://mirrors.saswata.cc/archlinux/
https://mirrors.saswata.cc/archlinux/
https://mirror.fossindia.ovh/archlinux/

Singapore
-----
https://download.nus.edu.sg/mirror/archlinux/
http://sg.mirrors.cicku.me/archlinux/
https://sg.mirrors.cicku.me/archlinux/
http://mirror.aktkn.sg/archlinux/
https://mirror.aktkn.sg/archlinux/
http://mirror.guillaumea.fr/archlinux/
```

Country-wise mirror list

Disk Configuration

Now, you need to partition your disk. The archinstall has a neat mechanism to help you here.

On the main menu, select “Disk Partition”. Inside this, select “Partitioning”.

```
Press ? for help
> Partitioning
  ← Back
```

Select Partitioning option

Here, use the option “Use a best-effort default partition layout”.

```

Disk configuration type
> Use a best-effort default partition layout
Manual Partitioning
Pre-mounted configuration

```

Best-effort partitioning

In the next dialog, use the TAB key to select your hard disk device and press the ENTER key.

```

Model          | Path      | Type | Size  | Free space | Sector size | Read only
-----
> [x] ATA VBOX HARDDISK | /dev/sda | scsi | 25 GiB | 25599 | 512 | False

```

Select Disk

Choose a partition type. Here, I am going with BTRFS partition. You can pick EXT4, a very well-tested file system, or XFS, f2fs etc.

```

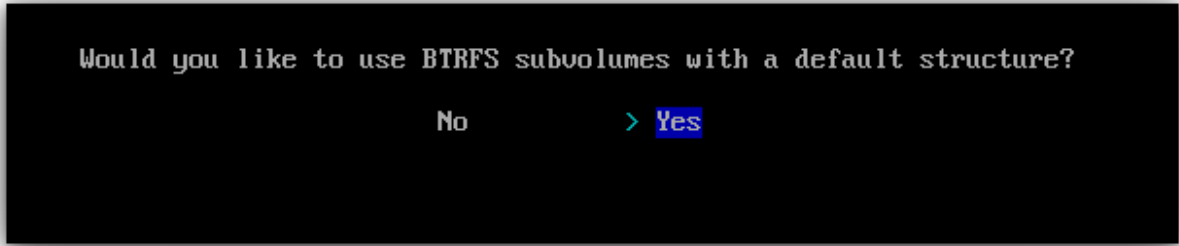
Filesystem
> btrfs
ext4
xfs
f2fs

```

Select BTRFS File System

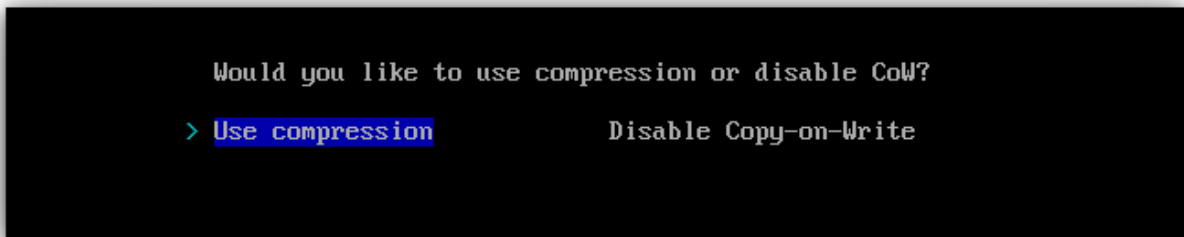
On the next screen, you will be asked to use a default subvolume structure or not.

Let's say you select "Yes".



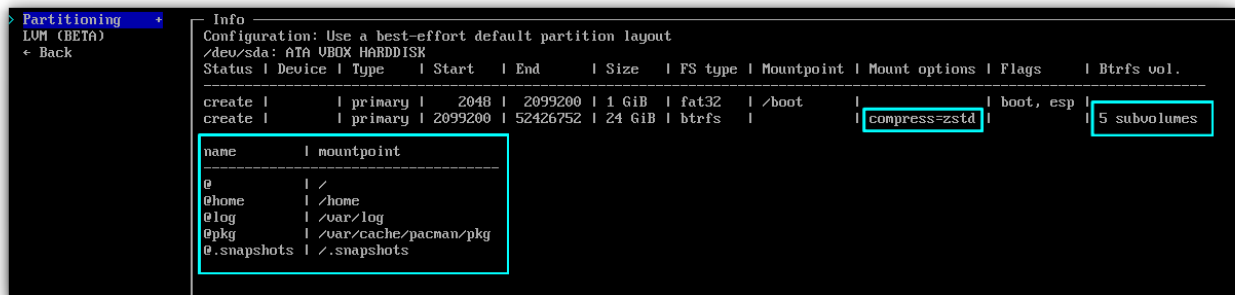
Create Subvolumes True

You will be asked to pick compression or disable copy-on-write. It is advised to select Compression, to enable a Zstd compression.



Use Compression option

This will create a partition for you, with subvolumes for /, /home, /var/log, /var/cache/pacman/pkg, and /.snapshots.

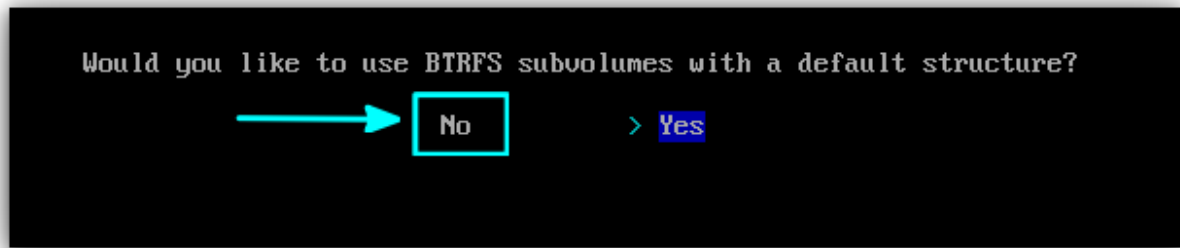


Subvolume listing (Click to expand the image)

Subvolumes are beneficial for users who want a granular control and use features like snapshots extensively.

If you are using a simple system, and not going to use such features, you can choose to avoid the subvolumes.

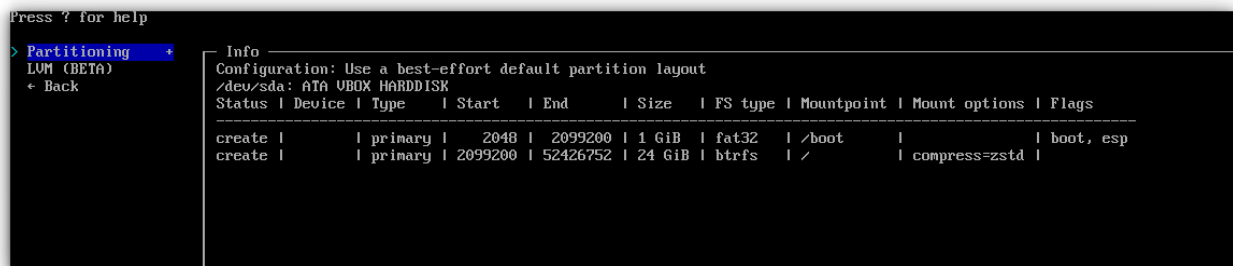
For this, pick "No" for BTRFS default subvolumes.



Subvolume choice

On the next screen, you should select "Use Compression" option.

Thus, you will get a simple partition for the system.



Simple no-subvolume partition.

Use the "Back" button to go to the installer main menu.

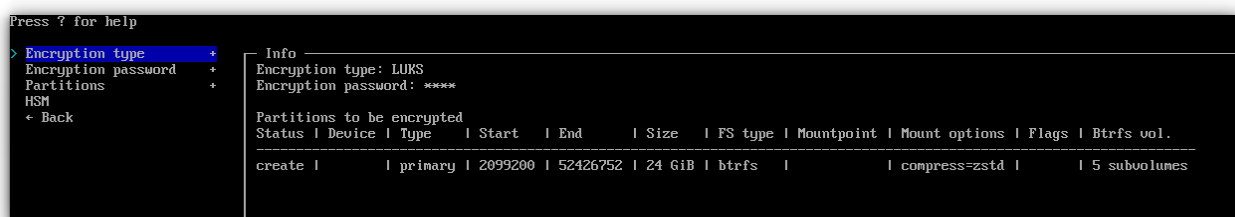
Disk Encryption

Disk encryption may introduce slight performance delay to the system. If your system is a casual home PC or an alternative system with no critical data, you can ignore the encryption.

Select the Disk Encryption option from the main menu. On the dialog box, select Encryption type and pick LUKS.

This will enable two other fields; Encryption password and Partition.

Fill the fields. Select the partitions that need to be encrypted using the TAB key.



Encryption overview (Click to expand the image)

Do not forget the encryption password. If you do, you'll lose access to the data on disk and formatting the entire operating system will be the only option for you.

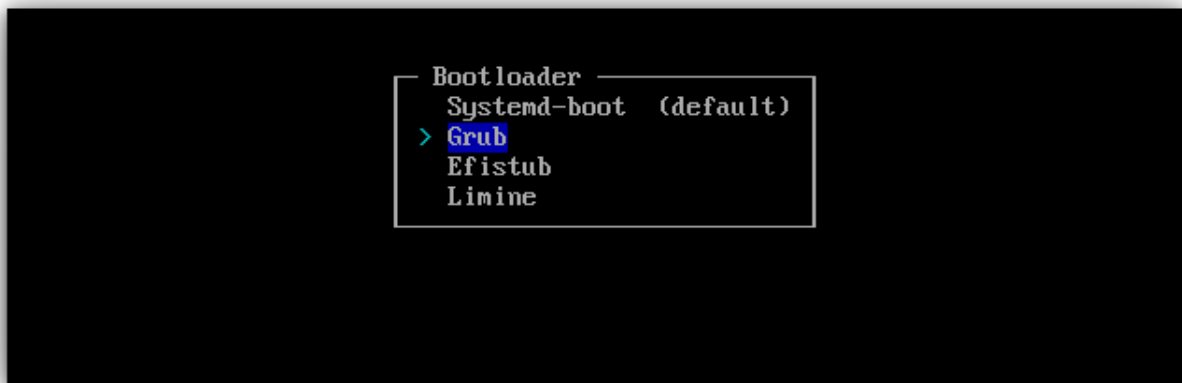
Swap

Swap on zram is enabled by default in the installer. If needed, you can disable it.

Bootloader

By default, it is set to systemd-boot. This is a simple bootloader for those who expect simplicity.

If you require familiar functionality, go for GRUB bootloader.



Select Grub Bootloader

Hostname

You can configure hostname here. By default, it is archlinux.

Root password

Next is Root password. Select it using enter key. Then enter and confirm a strong root password.

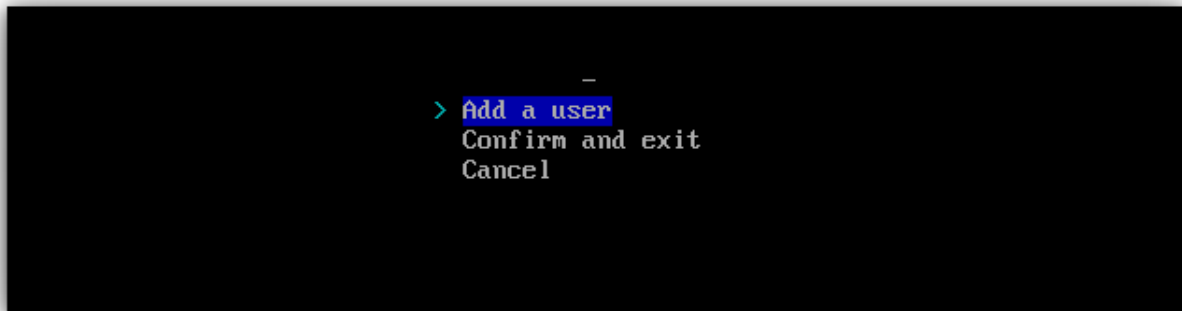


Root Password Setting

User creation

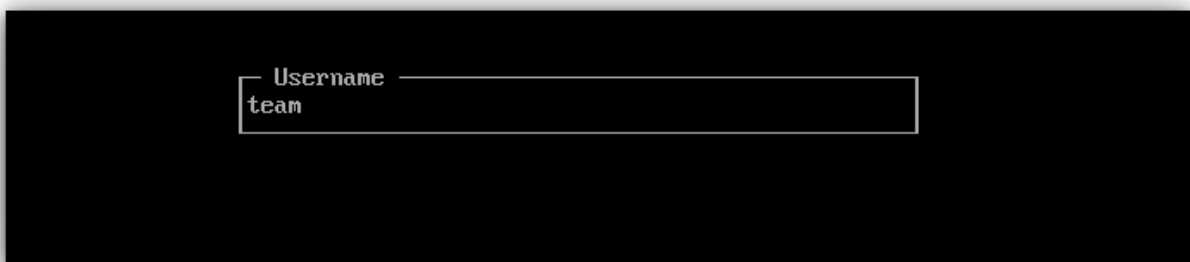
It is important to create a regular user account other than root account. This is for day-to-day purposes.

On User section, select "Add a user" option.



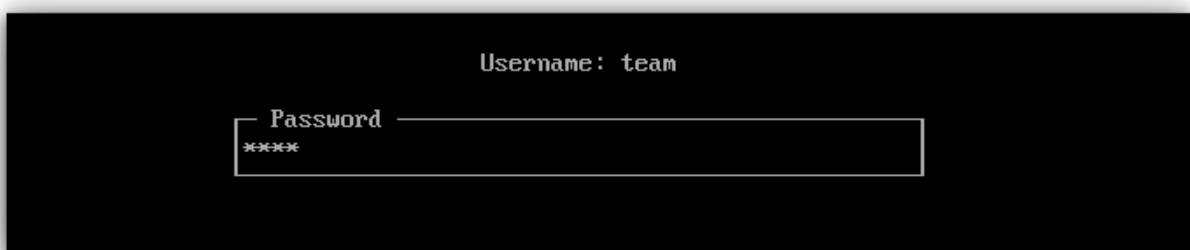
Click on "Add a user"

Here, enter the username.



Enter username

Now, enter a password.



Password for user

Confirm it by entering again when prompted. You will be asked whether the user a superuser or not.

Make the created user superuser (administrative privileges) by selecting the "Yes" option.

```
Username: team
Password: ****

Should "team" be a superuser (sudo)?

No          > Yes
```

Admin privileges to regular user

Now, use the "Confirm and exit" option.

```
username | password | sudo
-----|-----|-----
team     | ****              | True

Add a user
> Confirm and exit
Cancel
```

Exit user creation

Profile (Desktop selection)

The "Profile" field in the installer is where we will set desktop environments.

Select **Profile** → **Type**. Here, select the **Desktop** option.

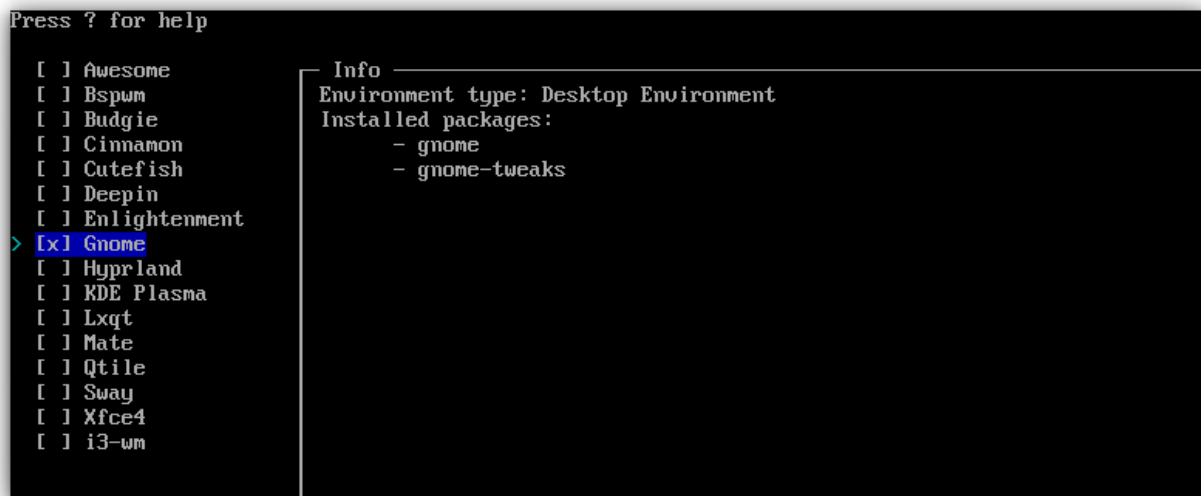
```
This is a list of pre-programmed default_profiles
```

```
┌ Main profile ─┐
> Desktop
Minimal
Server
Xorg
```

Select Desktop Option

On the next screen, select a desktop (desktops) using the TAB key and press enter.

Try to avoid installing multiple heavy desktops in one system. Like KDE Plasma and GNOME in one system is not recommended.

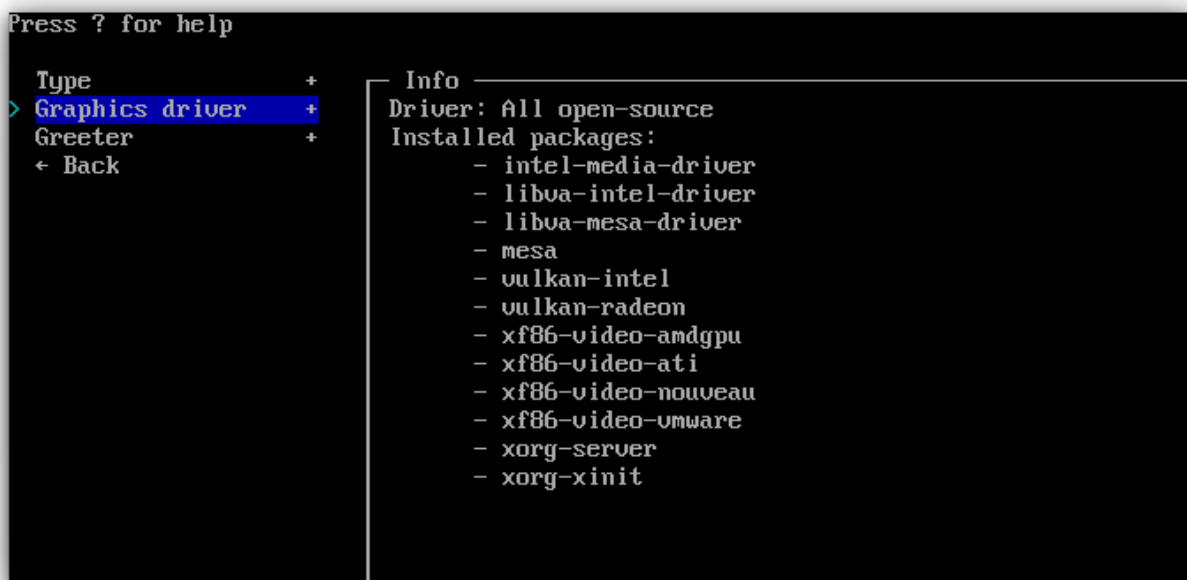


Select GNOME Desktop

You can choose one desktop like GNOME/Plasma and then choose one tiling window manager, making it install two desktop options.

Selecting a desktop and pressing enter will bring you to the driver selection settings.

For the test system, the installer automatically assigned all open-source drivers.



Driver packages

You can enter the “Graphics driver” settings and decide appropriate driver packs.

```

Press ? for help

AMD / ATI (open-source)
> All open-source (default)
Intel (open-source)
Nvidia (open kernel module for newer GPUs, Turing+)
Nvidia (open-source nouveau driver)
Nvidia (proprietary)
VMware / VirtualBox (open-source)

Info
-----
Installed packages:
- intel-media-driver
- libva-intel-driver
- libva-mesa-driver
- mesa
- vulkan-intel
- vulkan-radeon
- xf86-video-amdgpu
- xf86-video-ati
- xf86-video-nouveau
- xf86-video-vmware
- xorg-server
- xorg-xinit

```

Available drivers are listed

Normally, you should not be doing anything on the greeter, as it will be automatically selected (GDM for GNOME, SDDM for KDE Plasma etc.)

Audio settings

For Audio settings, you can select Pipewire or pulse audio.

```

Audio
-----
No audio server
> pipewire
pulseaudio

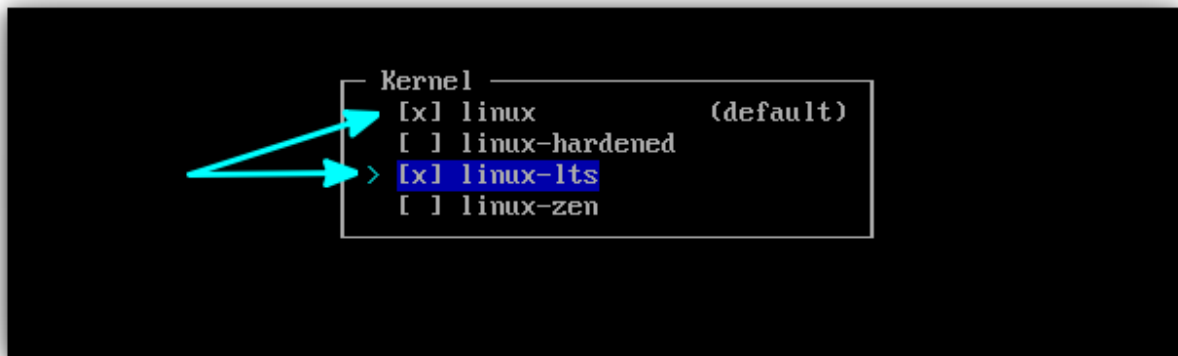
```

Select Pipewire

Kernel

You can either go with the default Linux kernel or select multiple kernels. Learn more about [kernel options in Arch Linux](#).

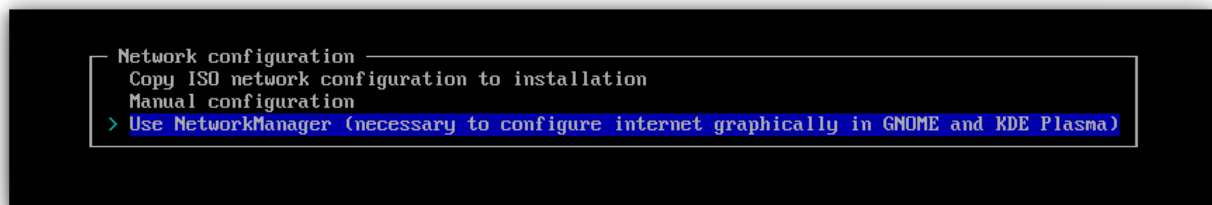
The screenshot below shows two kernels selected, `linux` and `linux-lts`.



Kernel selection

Network Configuration

In the Network Configuration settings, select "Use NetworkManager" option.



Use NetworkManager

Additional Packages

If you need to install additional packages to your system, you can do it at the installation stage itself.

Press enter key on "Additional package" option in main menu.

Now, just enter the proper name of the packages you want to install, separated with space.

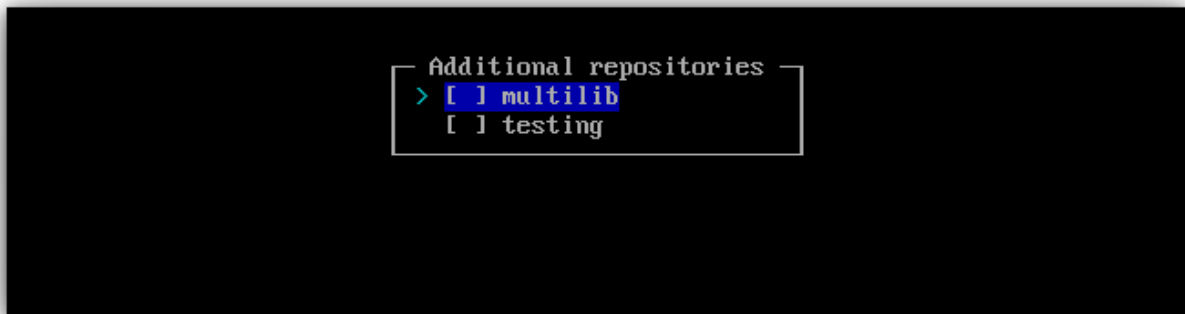
In the screenshot below, packages like firefox, htop, fastfetch, and starship are added.



Specify additional packages

Optional Repositories

You can enable `multilib` repositories using this setting. Select items using the TAB key and press enter. Learn about [various Arch repos here](#).



```
Additional repositories
> [ ] multilib
  [ ] testing
```

Additional Repositories

Timezone

Search and set the timezone based on your location. Asia/Kolkata for Indian Standard Time, US/Central for central timezone etc.

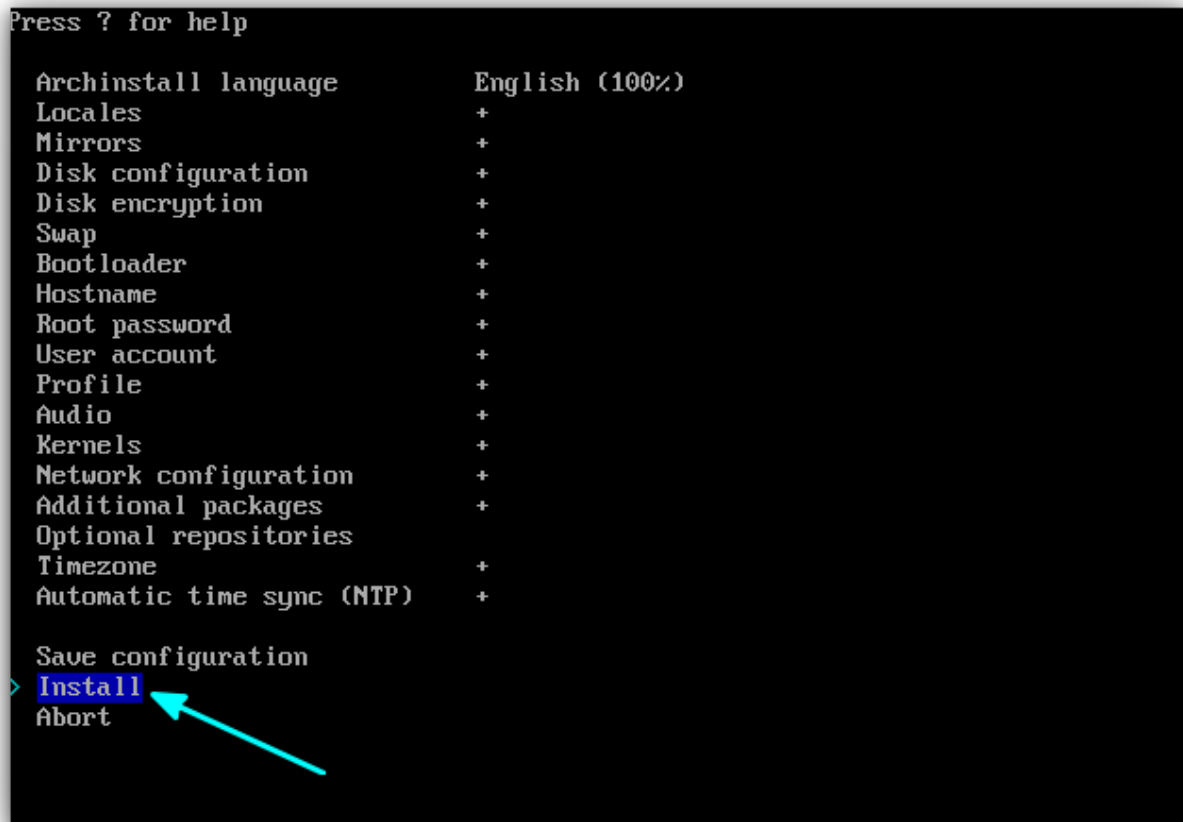
```
Timezone
Asia/Ashgabat
Asia/Ashkhabad
Asia/Atyrau
Asia/Baghdad
Asia/Bahrain
Asia/Baku
Asia/Bangkok
Asia/Barnaul
Asia/Beirut
Asia/Bishkek
Asia/Brunei
Asia/Calcutta
Asia/Chita
Asia/Choibalsan
Asia/Chongqing
Asia/Chungking
Asia/Colombo
Asia/Dacca
Asia/Damascus
Asia/Dhaka
Asia/Dili
Asia/Dubai
Asia/Dushanbe
Asia/Famagusta
Asia/Gaza
Asia/Harbin
Asia/Hebron
Asia/Ho_Chi_Minh
Asia/Hong_Kong
Asia/Hovd
Asia/Irkutsk
Asia/Istanbul
Asia/Jakarta
Asia/Jayapura
Asia/Jerusalem
Asia/Kabul
Asia/Kamchatka
Asia/Karachi
Asia/Kashgar
Asia/Kathmandu
Asia/Katmandu
Asia/Khandyga
> Asia/Kolkata
```

Timezone settings

Automatic Time Sync with NTP will be automatically enabled, and no need to change.

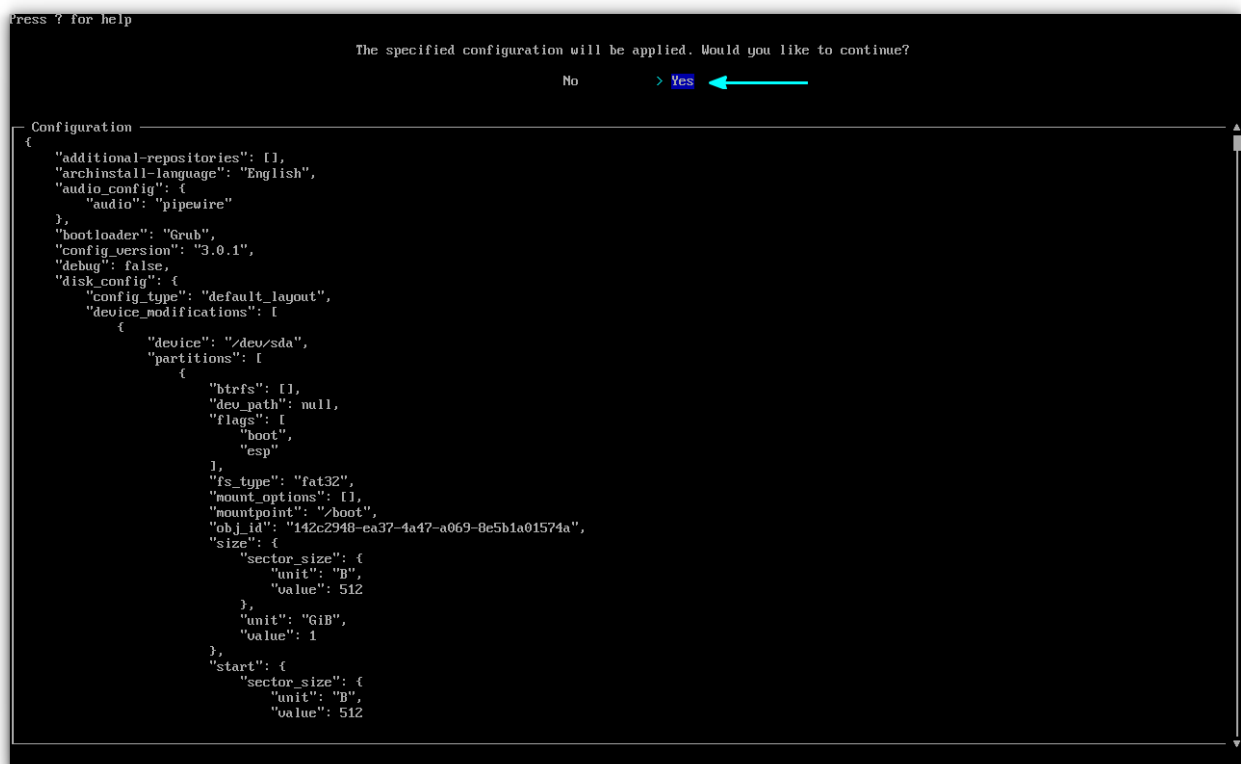
Start the actual install

Once all the settings have been done, you can use the **Install** option to start the installation procedure.



Use Install button

You will be asked to verify the installation configurations you have set. Once satisfied, enter on “Yes” option.



Confirm installation (Click to expand the image)

The process will be started, and you need to wait for some time to finish all the downloads and installations.

Step 6: Post Installation

Once the archinstall script finishes, it will ask you to chroot into the system for further settings. You can give NO to the question if you have nothing planned to do.

```
Would you like to chroot into the newly created installation and perform post-installation configuration?  
> No Yes
```

No chroot enter

You can now shut down the system.

shutdown now

```
Installation completed without any errors.  
Log files temporarily available at /var/log/archinstall.  
You may reboot when ready.  
archinstall 293.72s user 96.16s system 55% cpu 11:41.22 total  
root@archiso ~ # shutdown now
```

Shutdown the system

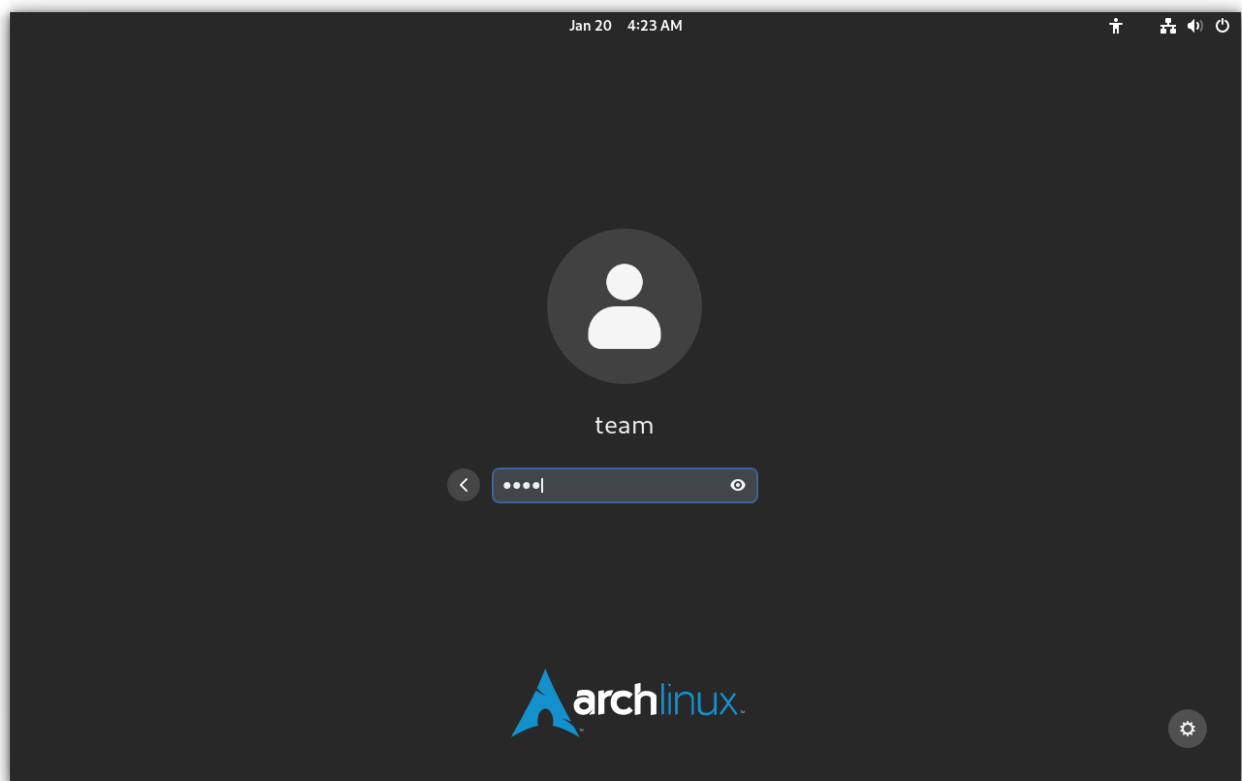
Once the system is shut down, remove the USB device from the port and boot the system.

This will bring you to the encryption page, if you have enabled encryption. Enter the password you have set.

```
A password is required to access the root volume:  
Enter passphrase for /dev/sda2: _
```

Enter encryption password

You will reach the login page. Enter the password to log in to your system.



Log in to the system

Enjoy Arch Linux with BTRFS and encrypted drive.